Konrad Trestka

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Education

University of Minnesota Twin Cities, College of Science and Engineering — Minneapolis, MN
B.S. Computer Science & B.S. Data Science — Expected May 2027
GPA: 3.75, Dean's List · Languages: English, Polish
Relevant Coursework: Data Structures & Algorithms, Machine Architecture & Organization, Optimization for Machine Learning, Applied Machine Learning, Operating Systems, Practice of Database Systems

Skills

Languages: Python, C/C++, Java, SQL, R, OCaml, JavaScript/TypeScript, Bash
Quant & Systems: event-driven systems, backtesting, market-data pipelines, risk controls (stop-losses, exposure caps, circuit breakers), Monte Carlo, time-series analysis
Tools & Libraries: pandas, NumPy, yfinance, Matplotlib, Alpaca API, Docker, Linux/Ubuntu, DigitalOcean, Git/GitHub

Quantitative & Trading Experience

1st Place — Quantitative Trading Competition — Apr 2026
Financial Mathematics Association — Judged by DRW & Cargill

- Placed 1st of all competing teams in a 10-day live algorithmic trading competition by engineering and deploying a fully automated, cloud-hosted (DigitalOcean) event-driven trading engine in Python on the Alpaca brokerage API.
- Developed a two-sleeve strategy capturing Post-Earnings Announcement Drift (PEAD) using intraday VWAP, 20-day SMA, and custom momentum scoring.
- Programmed strict capital-preservation controls — ATR-adjusted stop-losses, a 50% gross-exposure cap, and automated circuit breakers — achieving a 0.00% maximum drawdown in live forward-testing.
- Built asynchronous REST polling for 1-minute market data and Linux-based cron fail-safes to handle zero-volume stalls and API rate limits; presented and defended an After-Action Report to industry portfolio managers.

Undergraduate Student Researcher, Quantitative Derivatives — Jun 2026 – Present
University of Minnesota — Minneapolis, MN

- Designed a dual-lane options pricing system leveraging an Artificial Neural Network (ANN) as a fast surrogate for a 4-factor Markovian Path-Dependent Volatility (PDV) model.
- Engineered dynamic no-arbitrage SVI/SSVI guardrails that automatically revert the ANN to the baseline PDV model on unfamiliar or arbitrage-violating data.
- Executed Monte Carlo simulations on university supercomputers (HPC/SLURM) over large datasets of simulated S&P 500 market states.

Data Engineer Intern — Sep 2025 – Present
PURIS — Minneapolis, MN

- Architected automated ETL pipelines (Azure Data Factory → Azure SQL) processing daily REST/JSON payloads with idempotent upserts via stored procedures.
- Migrated production Docker container workloads to newly provisioned Ubuntu servers with verified backups and a staged, zero-loss cutover.

Projects

Algorithmic Trading Backtester
Python, pandas, yfinance, Matplotlib

- Built a scalable, object-oriented backtesting engine simulating algorithmic strategies over historical financial time-series, dynamically computing P&L and total equity.
- Engineered a data pipeline to ingest, clean, and structure market data into pandas DataFrames, with an optional news-sentiment sleeve to bias signals.

AJAP — Automated Job-Search Pipeline
Python, Flask, APScheduler, SQLite

- Self-hosted event-driven system that ingests roles from multiple sources, classifies them via an LLM API, and fires batched alerts; runs continuously as a scheduled Linux daemon.

Activities

Google Developer Student Club · Social Coding · Ski & Snowboarding Club